AGGLOMERATION PREMIUM IN POLYCENTRIC CITIES (SHORTENED VERSION)

BY
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Introduction

Urban land use theory since Alonso (1964) has mostly treated income as exogenous. The classical monocentric city allocates households in space through a tradeoff between commuting cost and land rent, and income only scales the bid–rent curve. This paper departs from that tradition by endogenizing income, which, in this model, is achieved through the agglomeration income premium. The premium is generated from the spatial clustering of firms. It is location–specific, varies continuously over a two–dimensional region, and is itself an equilibrium object.

The key technical choice to endogenize income is via a negative exponential distance–decay function, in the tradition of Fujita–Ogawa (1982). The distance-decay function is used to model population density and commuting cost. The advantage is conceptual and empirical: the function has favorable mathematical properties (smoothness, analytical tractability, etc.) and various urban gradients (including urban rent and density gradients) empirically follow exponential decay, according to the literature.

The model treats workers as nonatomic agents in a continuum. Each worker selects a location (x, y), the amount of land to consume, S, and the amount of the numeraire to consume, Z. They obtain Cobb–Douglas utility from consuming land and the numeraire. Workers face location–specific transport cost and earn agricultural income plus a location–specific premium $y_p(x, y)$ induced by firm clustering. Firms, in contrast, are discrete (atomic) agents. Firms must acquire discrete parcels of land and hire a fixed number of workers to generate revenue. Because the number of firms is finite, firm agglomerations have positive spatial footprint, i.e., they do not consume land on isolated points but on discrete plots. Firms compete with workers for land, but firms have no monopsony power: competition drives local wages and rents to their spatial equilibrium values.

Workers only vary in the wealth they have. Wealth is drawn from a Normal distribution. Sorting by wealth across space arises in the model. A spatial equilibrium in this model simultaneously determines: the map of firm agglomerations, the location–specific wage premium $y_p(x, y)$, the rent surface $r(x, y)$, the cost of commuting across space, and the vector of utility levels across space (which is determined by the wealth distribution). The paper concludes by exploring the spatial welfare properties of this spatial equilibrium, taking inspiration from Mirrlees (1972).

The contribution is conceptual: a tractable polycentric framework in which the aggressiveness of agglomeration forces is a structural parameter generating comparative statics over spatial inequality, density across space, and equilibrium welfare. Furthermore, it provides a framework that can be utilized in a structural model that explores the underlying dynamics responsible for various patterns of interest in transportation and land use, such as the finding that, in the postwar era, people suburbanized first and firms suburbanized afterwards (Steinnes 1977).

Related Literature

This paper builds off of previous literatures.

(i) Land use with Exogenous Income in Monocentric City Model. The canonical Alonso–Muth–Mills framework treats income as exogenous and studies how transport cost is capitalized into land rent (Alonso 1964; Muth 1969; Mills 1967). Later extensions incorporated heterogeneity in income and examined sorting in monocentric cities (Wheaton 1977; Glaeser 2004; Yinger 2018). Although these models do not allow for polycentricity and ignore the effect of city size and density on income, they are foundational to later spatial equilibrium models since they incorporate capitalization. These authors rightly noticed that the market for land is unlike other markets. Local goods and services, such as urban amenities, local government services, favorable climate, etc., can not be transported. The value of these local goods and services, if not paid through taxation, must get incorporated into land prices, an idea that was first brought to light by Oates (1969) when it was pointed out that better local government services should lead to higher house prices. This same idea is present in Alonso–Muth–Mills in a slightly different form, i.e., in transportation cost being capitalized into rent. More generally, the idea of spatial equilibrium involves equalizing utility (or, in the case of firms, profits) across space and having costs like rent and income adjust across space to ensure that this equalization happens.

(ii) Agglomeration Economies in Polycentric Cities using Distance-Decay Function. Fujita and Ogawa (1982) explicitly made firm location endogenous using a distance–decay formulation. Imai (1982) and Tabuchi (1986) studied how alternative spatial averaging measures capture the productivity consequences emanating from the level of firm clustering across space. Ideas from the Alonso-Muth-Mills model are borrowed, e.g., the bid-rent function deciding rent, but they apply to a more general spatial setting. One of the main tools Fujita and Ogawa (1982) use, which, due to its importance, is shown below, is the "locational potential" function:

$$F(x) = \int b(y)e^{-\alpha d(x,y)}dy \tag{1}$$

$F(x)$ is the locational potential at point x,[1] $b(y)$ is the density of businesses at point y, α is the distance-decay parameter, and $d(x, y)$ is the distance from point x to point y. In their model, firms calculate the locational potential of each region and use that knowledge to help them decide where to move. It is a measure of the agglomeration-economic benefit firms derive from each location. One thing to note is that firms derive agglomeration-economic benefit in their model from being around other firms. To make the model more concrete, refer to Figure 1 to see the shape of locational potential under different (hypothetical) values of the distance-decay parameter to get a general sense of the shape of the curvature that is being assumed:

(iii) Welfare and Utility Properties across Space. Mirrlees (1972) showed that curious results arise from a welfare maximization problem with a spatial component. More specifically, maximizing a Benthamite Social Welfare function would typically

[1] Note that x and y are both points on a 1 dimensional line. Their city is 1 dimensional.

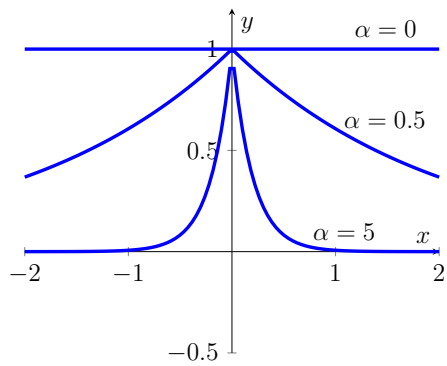

Figure 1: Distance Decay Parameter Effects

entail an equalization of endowments (and, as a result, utility) across agents, but, adopting a spatial framework similar to the Alonso-Muth-Mills model, Mirrlees finds that maximizing social welfare leads to the "unequal treatment of equals" where an equal distribution of wealth across individuals does not result. Wildasin (1986) showed that the outcome results from the fact that marginal utility of income depends on both prices and income. Since there are differences in rent across space (where rent is decided through the bid-rent process), Mirrlees (1972) model is one where consumers, even when maximizing utility, vary in how efficient they are at converting wealth into utility (this efficiency depends on their distance from the CBD). Therefore, when locational factors are considered, an optimal allocation of resources would result in a social-welfare optimum with varying levels of utility across individuals. This is purely a result of the spatial component of land consumption, which introduces a convexity into the social planner problem of how to find a distribution of workers across space which maximizes welfare.[2]

Summary: This paper bridges all these traditions while making appropriate changes: income premia in this model are constructed from clustering patterns across space. The wage gradient is *not* exogenous—it is an emergent spatial object. At the same time, the model preserves the land-rent bidding logic of the monocentric tradition, nesting the classical monocentric equilibrium as a special case. The modeling device that makes this synthesis tractable is a distance-decay function that operates across two-dimensional space. One key difference from Fujita-Ogawa (1982) is that agglomeration economies in this model results from matching, i.e., thick-labor-market, benefits that firms gain from locating close to employees, rather than the benefits from locating close to other firms. One motivation for this modeling choice is to have "jobs follow people" rather than "people follow jobs," a finding which is in line with post-war suburbanization in the U.S. (Steinnes 1977). Finally, the welfare implications of the sorting of workers by wealth level across space is explored. The elasticity of wealth on utility, and the

[2]It should be noted that a standard assumption in the literature is that land is a normal good. Rises in income will entail more land consumption. Mirrlees (1972) incorporates this standard assumption into his model.

properties that can be gleaned from it, is then compared to scenario of Mirrlees (1972).

Model Environment

Time is static. Space is a closed rectangle $[0, \bar{x}] \times [0, \bar{y}]$. This is the maximal feasible landmass. A subset of this region will be endogenously occupied in equilibrium.

Agents. There are three types of agents, the last of which plays a subsidiary role to the first two:

1. **Workers**: a nonatomic continuum. Workers choose (x, y) and land consumption S. Upon choosing a point in space to locate to, i.e., a set of coordinates (x, y), workers can then choose any amount of land to consume within that point. This means they exist in a single point rather than in a discrete lot in space. Land consumption is treated like any ordinary consumer good within that point. Workers each draw wealth ω from a Normal distribution $\mathcal{N}(\mu, \sigma^2)$. Wealth heterogeneity plays no role in production, but it affects land bids and, in spatial equilibrium, it affects the amount of land consumed in each (x, y)-point which affects the density levels that result across space. Wealth is the sole mechanism differentiating workers, i.e., ω defines a worker type for otherwise identical workers.

2. **Firms**: K discrete atomic agents. Each firm must acquire a fixed land-consumption value, S_b, and hire a fixed L_B number of workers to generate revenue. They generate no additional revenue from hiring more workers or owning more land. Even though firms are discrete, their hiring is done over small circular parcels with area equal to ϵ_K (in order for employment to be measurable within integrals, firm hiring cannot be done on a discrete set of points that lack two dimensional area). Furthermore, if two circular parcels of size ϵ_K touch on the two dimensional plane, they automatically agglomerate and form a circle of double the area. If N firms agglomerate, their joint land footprint is a single circular parcel of area $N\epsilon_K$, with radius $\sqrt{N\epsilon_K/\pi}$.

3. **Landlords**: They do not play a very large role. Their role is to ensure that the market for renting land is competitive. They will rent out land to the workers/firms who bid the highest price for it. Firms get priority because it is assumed that they are always able to bid a higher price than workers are (this notion is formalized later on).

Worker/Consumer Utility Maximization Problem

Workers earn:

$$y_T(x, y) = y_a + y_p(x, y) \tag{2}$$

where:

$y_T(x,y)$: The total income a worker expects to earn by locating in (x,y),
y_a : Agricultural income, i.e., income that does not depend on (x,y).
$y_p(x,y)$: location-based agglomeration premium workers expect to make when living in location (x,y).

Workers solve:

$$\text{Max}_{Z,S} \;\; U = Z^{\alpha} S^{\beta} \tag{3}$$

$$\text{subject to} : r(x,y)S + Z + \tilde{d}(x,y) = \omega + y_T(x,y) \tag{4}$$

where:

$\tilde{d}(x,y)$: location-specific commuting cost,
$r(x,y)$: location-specific rent for one unit of land.

It should be noted that the problem above is not a spatial problem. Consumers solve the problem above for every (x,y) point before choosing where to locate.

Firm Profit Maximization Problem (a Spatial Problem)

$$\text{Max}_{x,y} \;\; \pi(x,y) = R\, f(S_b, L_b) + \gamma F(x,y) - r(x,y)S_b - (y_a + \tilde{y_p}(x,y))L_b \tag{5}$$

where:

R : Revenue per product produced,
$f(S_b, L_b)$: fixed level of production that results when firms hire L_B workers and rent S_B land,
γ : location-based agglomeration premium firms receive due to better matching,
$F(x,y)$: a measure of worker population density within commuting distance of (x,y)
$\tilde{y_p}(x,y)$: location-based agglomeration premium firms expect to pay to each worker when located in location (x,y).

It should be noted that because the labor and land that firms need to buy in order to produce is fixed, the problem above is a spatial problem. The only choice firms make is which (x,y) point to locate to.

Bid Rent

The worker/consumer utility maximization problem has a solution for each location (x,y). The solution to that problem will vary by wealth ω. Let $Z^*(x,y;\omega)$ and $S^*(x,y;\omega)$ denote the solution for a specific location and wealth level.

Furthermore, as will be shown in the next section, a spatial equilibrium is

imposed which means that for a fixed wealth type ω, there is equal utility across all locations actually occupied by type ω workers. Let $\bar{U}_\omega$ denote this equilibrium utility level. Then, under spatial equilibrium:

$$U(Z^*(x, y; \omega), S^*(x, y; \omega)) = \bar{U}_\omega.$$

To construct the bid-rent function, note that workers have the following budget-set (as shown in the utility maximization problem):

$$r(x, y)S + Z + \tilde{d}(x, y) = \omega + y_a + y_p(x, y)$$

The expression above can be simplified by having y_a subsumed within ω. Since y_a is not location-dependent, this can easily be accomplished with no effect on how the model is interpreted. Therefore, removing y_a and rewriting the expression above with rent on the LHS leads to the following:

$$r(x, y) = \frac{\omega + y_p(x, y) - \tilde{d}(x, y) - Z}{S}$$

Then, workers of type ω in (x, y) will bid up rent with the restriction that they will never let their utility levels fall below $\bar{U}_\omega$:

$$r(x, y; \omega) = \max_{Z,S} \left\{ \frac{\omega + y_p(x, y) - \tilde{d}(x, y) - Z}{S} \;\middle|\; U(Z, S) \geq \bar{U}_\omega \right\} \tag{6}$$

However, since workers must consume optimally according to the consumer utility maximization problem, the problem above can be reformulated as:

$$r(x, y; \omega) = \max_{Z^*,S^*} \left\{ \frac{\omega + y_p(x, y) - \tilde{d}(x, y) - Z^*}{S^*} \;\middle|\; U(Z^*, S^*) = \bar{U}_\omega \right\} \tag{7}$$

The more important question is how do landlord make their choice of which consumers to rent the land to. The optimal bundle that workers consume, (Z^*, S^*), can be rewritten to be a function of ω, $y_p(x, y)$, $r(x, y)$, and $\tilde{d}(x, y)$. Notice that all workers face the same choices of $y_p(x, y)$, $r(x, y)$, and $\tilde{d}(x, y)$. So, the choice landlords face is which ω will give them the highest rent. Reformulating the bid-rent function (using various steps shown in the main paper) leads to the following function which demonstrates the landlord's rent maximization problem:

$$\text{Max}_\omega \;\; \left(\frac{(\frac{\beta}{\alpha})(\omega + y_p(x, y) - \tilde{d}(x, y))^{1+\frac{\alpha}{\beta}}}{(\bar{U}_\omega)^{\frac{1}{\beta}}(1 + \frac{\beta}{\alpha})^{1+\frac{\alpha}{\beta}}}\right) \tag{8}$$

The only remaining question is how firms rent land. The firm profit maximization problem has the term $R\ f(S_b, L_b)$. It is assumed that this term is set to a level high enough to allow firms to outbid consumers of any ω level. Therefore, firms do not need to be included in the bid-rent function. They will accept the rent values that

result from consumers bidding up rent. It will be assumed that firms have priority on whether to rent land at any location, and, if they choose not to, consumers are allowed to rent at that location. Unlike consumers, there is only one type of firm, and it is assumed that they are marginal enough players in the land market that they do not compete with each other for land in a manner that causes rent to rise to the maximum value firms are able to pay. It is only consumers who are willing to bid rent up to the maximum.

Spatial Equilibrium

A spatial equilibrium is defined as a set of $(r(\cdot),\ y_p(\cdot),\ \tilde{y_p}(\cdot),\ \tilde{d}(\cdot),\ F(\cdot))$ functions and an assignment of workers and firms to locations such that the following conditions hold:

1. **Worker optimality:** for each wealth type ω located in (x, y), the (Z, S) values consumed solves the utility maximization problem.

2. **Worker spatial optimality:** All locations occupied by type ω workers yield the same $\bar{U}_\omega$ where $\bar{U}_\omega$ represents the highest utility that workers of type ω can attain.

3. **Firm spatial optimality:** Each firm is located on some (x, y) (and does hiring on a circular parcel containing (x, y)) such that no firm has any incentive to relocate, and all firms make the same level of profit.

4. **Land market clearing:** at each (x, y), the rent equals the highest bid, which is decided by ω:
$$r(x, y) = \max_{\omega}\{r(x, y; \omega)\}.$$

Additional Model Equations and Relationships

The following equations are also part of the model. Their form may look complicated at first, but they can be intuited more easily by realizing that they each represent weighted averages across space of the form:

$$H(x, y) = \frac{\int_y (\int_x h(x, y)\ w_{x,y}\ dx) dy}{\int_y (\int_x w_{x,y}\ dx) dy} \tag{9}$$

where $H(x, y)$ represents the spatially weighted average of function $h(x, y)$ and $w_{x,y}$ represents the spatially varying weights. No specific formula is used universally as weights for all such functions in this paper. The formula for the weights will depend on the function that is being averaged across space. The reason why spatial averages are needed is because the values given to $(r(x, y),\ y_p(x, y),\ \tilde{y_p}(x, y),\ \tilde{d}(x, y),\ F(x, y))$ are not decided solely by what is happening on the isolated (x, y) points themselves, but also by what is happening in their vicinity.

Before listing the spatially-weighted functions, density needs to be defined. Let $\ddot{S}_{x,y} = S^*(x, y; \omega)$ for the ω-type worker who outbids all other worker types in

location (x, y). More intuitively, it is the lot size that actually gets purchased in (x, y). Furthermore, consider a curve defined by taking the set of (x, y) values that a ω-type worker lives in.[3] Let c_ω be the length of all the curves in the level set produced by workers of type ω.

In order for density to be defined, it is necessary to account for how the mass of workers living within the ω-level-set is not necessarily accurately represented by comparing c_ω to the length of $c_{\omega'}$ for some other ω' worker-type. Relative population sizes of different worker types can only be found by the PDF values of $\mathcal{N}(\mu, \sigma^2)$. To make this idea more concrete, consider Figure 1 below:

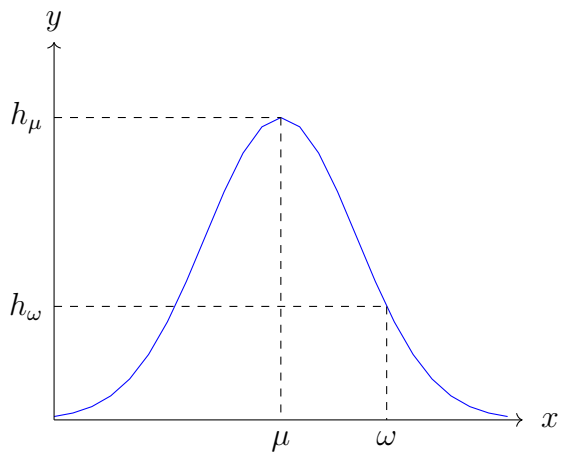

Figure 2: Wealth Distribution

The h_μ and h_ω values correspond to the height of the PDF at wealth levels μ and ω. The ratio of population sizes for workers of these two types is h_ω/h_μ, which means that if, for example, $h_\omega/h_\mu = .5$, then there are half as many workers of type ω as of type μ. With that in mind, $\delta_{x,y}$ is defined as follows:

$$\delta_{x,y} = \frac{c_\mu}{c_{\omega_{x,y}}} \cdot \frac{h_{\omega_{x,y}}}{h_\mu} \tag{10}$$

where $\omega_{x,y}$ is the wealth level that lives in location (x, y). Therefore, $\delta_{x,y}$ represents the scaling factor that $c_{\omega_{x,y}}$ needs to be multiplied by to ensure $c_{\omega_{x,y}}$ is proportional to the size of the population living there. The scaling is done relative to the c_μ curve. With that in mind, consider the following: $\delta_{x,y}\ddot{S}_{x,y}$. It is equal to the amount of land consumed in (x, y) multiplied by the scalar that is used to make the level-sets proportional to the population living there. The measure of density used in this paper utilizes this scaling process:

[3]The following condition is sufficient to ensure that no level-set has positive area: $\nabla y_p(x, y) - \nabla \tilde{d}(x, y) - \ddot{S}_{x,y} \nabla r(x, y) \neq \mathbf{0}$. In words, it means that for all points (x, y), there must be marginal change in at least one direction for which a change in the agglomeration premium is not entirely canceled out by the simultaneous change in transportation and total rent payments that results. If such a condition holds, then every level-set will be a set of (potentially non-connected) 1-dimensional curves. None will have positive area.

$$\phi(x, y) = \frac{1}{\delta_{x,y} \ddot{S}_{x,y}} \tag{11}$$

$\phi(x, y)$ represents the density of households at point (x, y). The density measure is the inverse of scaled lot sizes. If lot sizes are larger, density is lower. Scaling is needed because the level set that (x, y) belongs to might have a higher (or lower) share of the population than can be gleaned by its size on the 2-dimensional grid alone.

The distance function is defined as follows:

$$d_{(x',y')}(x, y) = \sqrt{(x - x')^2 + (y - y')^2} \tag{12}$$

The following represents the density-scaled transportation cost function:

$$\ddot{d}_{(x',y')}(x, y) = d_{(x',y')}(x, y)\ k \int_0^1 \frac{1}{\phi(\bar{x}(t), \bar{y}(t))}\ dt \tag{13}$$

where:

$$\begin{aligned} k &: \text{ transportation cost per unit of distance,} \\ \bar{x}(t) &: \text{ x-value of the parametric path going from } (x, y) \text{ to } (x', y'), \\ \bar{y}(t) &: \text{ y-value of the parametric path going from } (x, y) \text{ to } (x', y') \end{aligned}$$

Therefore, $\ddot{d}_{(x',y')}(x, y)$ represents the density-scaled cost of of transporting from (x, y) to (x', y'). The integral measures the inverse density along the parametric path that goes from (x, y) to (x', y'). That means that if the straight path from (x, y) to (x', y') involves going through low density areas, the transportation cost will be higher. If the straight path from (x, y) to (x', y') involves going through high density areas, the transportation cost will be lower.[4] The reason why transportation cost is scaled by the inverse of density is because there needs to be some way of accounting for the fact that although this model has people located on individual (x, y) points, in a real land market, larger discrete lots would require more land which would take longer to drive through.

Using the density-scaled transportation cost function, let $\ddot{d}_A(x', y')$ denote the average density-scaled transportation cost of getting to a firm for a household located in (x', y'):

$$\ddot{d}_A(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} i_{x,y}\ \ddot{d}_{(x',y')}(x, y)\ dx\ dy}{\int_0^{\bar{x}} \int_0^{\bar{y}} i_{x,y}\ dx\ dy} \tag{14}$$

[4]Note that if no additional inconvenience was imposed, it would be better to go through the cost-reducing parametric path rather than a parametric path representing a straight line from (x, y) to (x', y'). The reason why this is not done is because restrictive conditions would have to be placed on the shape of $\phi(\cdot)$ to come up with an analytical solution for the cost-reducing path. Even though $\phi(\cdot)$ is continuous, there are no restrictions placed on its movements across space such that a cost-minimizing parametric path can be given a neat analytical expression. There are algorithms, such as Dijkstra's algorithm that would be able to approximate it in a computational setting, but, for the sake of analytical clarity, this model assumes that the parametric path is a straight line.

where $i_{x,y}$ is an indicator for whether (x, y) is in one of the circular parcels in which firm employment happens. Expression (14) takes the form of (9) where $i_{x,y}$ acts as the weights. $\ddot{d}_A(x', y')$ is used to define one of the distance-decay parameter terms:

$$\psi_{x',y'} = \frac{\upsilon}{\ddot{d}_A(x', y')} \tag{15}$$

$\psi_{x',y'}$ represents the inverse of the average density-scaled transportation cost of getting to a firm for a household located in (x', y') (scaled by some value υ). Recall the distance decay parameter in Figure 1. $\psi_{x',y'}$ is going to be the density decay parameter in the expression for $\tilde{d}(x', y')$, which is the average transportation cost (to commute to work) for workers located in (x', y'). Note that $\ddot{d}_A(x', y')$ does not represent the average transportation cost to commute to work. Why is this? It is because workers, when looking for a job, have various reasons to end up working in closer locations over those further away. Distance is one of those reasons, but not the only one. There are search-and-match frictions across space, and workers need to ensure a successful match with a firm to end up working there. If the labor market were to be modeled in all it intricate detail, complicated search and matching dynamics would need to be introduced. This model avoids these dynamics through the use of a distance-decay parameter to scale values of $\tilde{d}(x', y')$. Whereas $\ddot{d}_A(x', y')$ represents the average (density-scaled) distance to all firms for workers living in location (x', y'), the $\tilde{d}(x', y')$ measure represents the (density-scaled) distance to commute to firms such that closer firms are weighted more heavily than those further away:

$$\tilde{d}(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} i_{x,y}\ \ddot{d}_{(x',y')}(x, y)\ e^{-\psi_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy}{A_{x',y'}} \tag{16}$$

where:

$$A_{x',y'} = \int_0^{\bar{x}} \int_0^{\bar{y}} i_{x,y}\ \ e^{-\psi_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy \tag{17}$$

$\tilde{d}(x', y')$ is calculated using the form in (9) with weights equal to $i_{x,y}\ e^{-\psi_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}$. These weights correspond to the negative exponential with $\psi_{x',y'}$ as the distance-decay parameter and the density-scaled transportation cost function as the distance function. Furthermore, the $i_{x,y}$ ensures that the average is only calculated using locations from which firms are employing workers. The overall form is similar to the Fujita-Ogawa construction with one notable difference: the distance-decay parameter value varies across space. If (x, y) is a location where the cost of commuting to firms is on average lower, it is a location with higher than average firm density. Higher firm density nearby means a larger likelihood of being matched to a firm nearby. This means that workers will treat (x, y) as a location that they can be expected to commute more cheaply from if they choose to live there. Figure 1 shows that a higher distance-decay parameter means that geographically closer points count for more in the average. The same idea holds here. If there is higher firm density close to (x, y), then commuting will be cheaper on average from point (x, y), thereby making $\ddot{d}_A(x', y')$ lower and $\psi_{x',y'}$ higher.

Therefore, $\tilde{d}(x', y')$ represents the expected cost of commuting to a firm for a worker located in (x', y') when such a cost has been transformed to fit the form of a negative exponential function in the manner of Fujita-Ogawa (1982).

The agglomeration income premium that workers receive is calculated in a similar manner, by taking the average across space of the agglomeration income premium that firms offer using the negative exponential function with $\psi_{x',y'}$ as the distance decay parameter:

$$y_p(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} i_{x,y}\ \tilde{y}_p(x, y)\ e^{-\psi_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy}{A_{x',y'}} \tag{18}$$

The role of the distance decay parameter is the same as in the calculation of $\tilde{d}(x', y')$. The intuition behind (18) is that households can observe (by looking at vacancies and advertised wages across space) what the agglomeration premium is on average across space, even if they can not predict with certainty what firm they will be matched to. This model assumes that the agglomeration premium workers make is equal to this spatial average.[5]

Firms go through a similar process. First, they must come up with a measure for how far workers live from any location (x', y'):

$$\ddot{d}_B(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} (1 - i_{x,y})\ \ddot{d}_{(x',y')}(x, y)\ dx\ dy}{\int_0^{\bar{x}} \int_0^{\bar{y}} (1 - i_{x,y})\ dx\ dy} \tag{19}$$

where $(1 - i_{x,y})$ is an indicator for whether (x, y) is land in which a household, rather than a firm, has located. Using $\ddot{d}_B(x', y')$ defined above, a new distance-decay parameter can be defined. This is the distance-decay parameter that firms will use:

$$\lambda_{x',y'} = \frac{\tau}{\ddot{d}_B(x', y')} \tag{20}$$

where τ is some scalar. The agglomeration premium to income that firms pay their workers will be a spatial average of the agglomeration premium that workers in any location (x, y) receive (similar to workers, they do not know what workers they will be matched to, but they can observe wages across space). The weighting uses the negative exponential function with a distance-decay parameter of $\lambda_{x',y'}$:

$$\tilde{y_p}(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} (1 - i_{x,y})\ y_p(x, y)\ e^{-\lambda_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy}{B_{x',y'}} \tag{21}$$

where:

$$B_{x',y'} = \int_0^{\bar{x}} \int_0^{\bar{y}} (1 - i_{x,y})\ \ e^{-\lambda_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy \tag{22}$$

[5]One way of conceiving this is that the model is trying to represent that people will regularly switch jobs after moving to a location and there is uncertainty around which matches will be successful. Instead of creating a stochastic model with complicated probability expressions, this model deals with the spatial uncertainty by taking a spatial average across space and sets the worker's realized wage to that average.

The expressions given to $y_p(x', y')$ and $\tilde{y_p}(x', y')$, taken in unison, imply that they are simultaneously determined by each other. This is expressing the idea that firms and workers both use spatially-weighted averages of the agglomeration premium value that they observe the other party paying/requesting (across space) to decide what they want to pay/request. In the end every worker gets matched to a firm, and the spatial configuration that emerges leads to the realization of firms (workers) paying (receiving) the spatially-weighted values they observed workers (firms) around them receiving (paying).

Finally, $F(x', y')$ represents the premium that firms receive from better matches in areas with a higher density of workers. To get this value, a distance-decay expression is taken using the population density measure, $\phi(\cdot)$:

$$F(x', y') = \frac{\int_0^{\bar{x}} \int_0^{\bar{y}} (1 - i_{x,y})\ \phi(x, y)\ e^{-\lambda_{x',y'}\ \ddot{d}_{(x',y')}(x,y)}\ dx\ dy}{B_{x',y'}} \tag{23}$$

It is worth reiterating that one major difference between the framework employed here and in Fujita-Ogawa (1982) is that the framework employed here assumes firms derive benefits from being around a higher concentration of workers rather than a higher concentration of firms.

Properties under Spatial Equilibrium

The following properties must hold in Spatial Equilibrium. For marginal changes in the x and y direction, higher (lower) costs associated with the new location, i.e., higher (lower) transportation costs and/or higher (lower) rent, must be balanced out by more (less) benefits in the form of higher (lower) land consumption, higher (lower) numeraire consumption, and/or a higher (lower) agglomeration income premium. The land consumption and numeraire consumption values refer to the values that would be consumed in location (x, y) by the wealth type that would be living there in the spatial equilibrium. Note that marginal changes in any direction may involve changes in the wealth-type that occupy that location (in equilibrium).

$$\frac{\partial y_p(x, y)}{\partial x} = \frac{\partial r(x, y)}{\partial x} S^*(x, y) + r(x, y) \frac{\partial S^*(x, y)}{\partial x} + \frac{\partial Z^*(x, y)}{\partial x} + \frac{\partial \tilde{d}(x, y)}{\partial x} \tag{24}$$

$$\frac{\partial y_p(x, y)}{\partial y} = \frac{\partial r(x, y)}{\partial y} S^*(x, y) + r(x, y) \frac{\partial S^*(x, y)}{\partial y} + \frac{\partial Z^*(x, y)}{\partial y} + \frac{\partial \tilde{d}(x, y)}{\partial y} \tag{25}$$

Similarly, it must be the case that for firms, any marginal changes in the x or y direction is such that the marginal changes to the matching premium is canceled out by marginal changes in their costs, i.e., the rent they pay for land and the agglomeration premium they pay to workers:

$$\gamma \frac{\partial F(x, y)}{\partial x} = \frac{\partial r(x, y)}{\partial x} S_b + \frac{\partial \tilde{y}_p(x, y)}{\partial x} L_b \tag{26}$$

$$\gamma \frac{\partial F(x,y)}{\partial y} = \frac{\partial r(x,y)}{\partial y} S_b + \frac{\partial \tilde{y}_p(x,y)}{\partial y} L_b \tag{27}$$

Moreover, the above four rules can be generalized so that they hold for a marginal change in any direction, not just in the x or y direction. Let ∇ represent a marginal change in any direction:

$$\nabla y_p = S^* \nabla r + r \nabla S^* + \nabla Z^* + \nabla \tilde{d}, \qquad \gamma \nabla F = S_b \nabla r + L_b \nabla \tilde{y}_p.$$

Welfare Analysis

The last contribution of this thesis is to explore a welfare property that results from the model. The marginal change in utility from a marginal change in wealth is as follows:

$$\frac{\partial \bar{U}_\omega}{\partial \omega} \cdot \frac{\omega}{\bar{U}_\omega} = \frac{(\beta + \alpha)}{(1 - \frac{c_{x,y}}{\omega})} \tag{28}$$

where $c_{x,y} = \tilde{d}(x,y) - y_p(x,y)$ can be thought of as the "locational cost" of living somewhere. A negative value implies a benefit, i.e., the agglomeration income premium is higher than the commuting cost. This is what would most likely hold for most people. However, for those on the higher end of the wealth distribution, it may be that they get enough of a benefit in the form of cheap land to commute a distance that leads to a commuting cost higher than their agglomeration income premium. Note that no corner solutions in the uility maximization problem implies that:

$$\omega = r(x,y)S^*(x,y) + c_{x,y} + Z^* \implies c_{x,y} < \omega \implies 0 < \frac{c_{x,y}}{\omega} < 1$$

Therefore, the elasticity of wealth on utility is always positive (as expected). Furthermore, it is lower for higher values of wealth. Unlike Mirrlees (1972), the transportation and agglomeration premium now have an effect on the elasticity. When the agglomeration premium is very high, workers will benefit less from receiving more wealth. The workers who will benefit the most from more wealth are those with low wealth, high transportation cost and a low agglomeration premium. These workers are living in low density areas but have to commute long distances to get to their jobs. They are more efficient at converting wealth into utility because they live further out in the periphery where land is cheaper. Workers in high density areas, including those with low wealth, will benefit less from wealth distribution because land is expensive in the high density areas they live in. So, they are less efficient at converting wealth into utility. While there are similarities to Mirrlees (1972), density's effect on transportation cost and income introduces a new dynamic that affects the social planner problem.

Future Steps and Possibilities Going Forward

The first task for making the model operational is to establish the existence of spatial equilibrium. Conceptually, this requires turning the system of objects:

$$\big(r(\cdot),\, y_p(\cdot),\, \tilde{y}_p(\cdot),\, \tilde{d}(\cdot),\, F(\cdot),\, \phi(\cdot),\, i(\cdot)\big)$$

into a fixed point of a well–defined operator. For a given parameter vector $\theta = (k, \gamma, \upsilon, \tau, \alpha, \beta, R)$ and a candidate profile of functions on $[0, \bar{x}] \times [0, \bar{y}]$, I need to define an equilibrium operator:

$$\mathcal{T}_\theta \colon \Xi \;\mapsto\; \Xi', \qquad \Xi = (\, r,\; y_p,\; \tilde{y}_p,\; \tilde{d},\; F,\; \phi,\; i\,).$$

and a spatial equilibrium corresponds to a fixed point Ξ such that $\mathcal{T}_\theta(\Xi) = \Xi^*$. The natural next step is to place these functions in an appropriate function space (for example, a bounded, closed, convex subset of $C([0, \bar{x}] \times [0, \bar{y}])$ or L^∞), verify continuity and compactness properties of $\mathcal{T}_\theta$, and then invoke a fixed point theorem to establish existence.

Once existence has been established, the next step is to study stability and multiplicity. For spatial models like these, there is no reason to expect uniqueness: different initial clustering patterns may converge to different spatial configurations. So, what would need to be done to explore the different spatial configurations is to create a dynamic simulation and test the spatial equilibrium that results from different initial conditions. One thing that should be explored is how altering k affects the equilibrium. This would be the post-war situation in the U.S. where the introduction of automobiles made suburbanization possible. It would be interesting to see if the simulation leads to the post-war patterns made known by Steinnes (1977) where "jobs follow people" in the process of suburbanization.

Once that is done, I want to see if the spatial model here can be turned into a structural model which gets calibrated on real data. I can look for data that allows me to identify density patterns across space for different megaregions across the globe. I would need to look for labor market data that is spatially granular enough for the purposes of this model. This step of the process has not been fleshed out in great detail, but it remains part of the eventual goal.

Conclusion

This paper generalizes the Alonso–Muth–Mills framework to a polycentric environment in which (i) workers form a nonatomic continuum, (ii) firms are discrete (atomic) and may co-locate into clusters, and (iii) wages include an endogenous, location–specific agglomeration premium generated by thick labor markets. Commuting costs are distance– and density–weighted via a line integral that internalizes the geometry of settlement and lot size choices. The model delivers: (a) wealth sorting along connected level sets of *locational cost* $c(x, y) = \tilde{d}(x, y) - y_p(x, y)$; (b) continuous rent and density surfaces; and (c) a tight welfare connection between the spatial share of agglomeration income and the elasticity of wealth into equilibrium utility.

Qualitatively, the framework allows for the monocentric city as a specific case while accommodating polycentric megaregions in which centripetal forces operate along multiple directions. Quantitatively, it provides a coherent way to compare equilibria under changes in commuting technology, firm clustering, and the strength of agglomeration.

Three limitations invite future work. First, while the $\delta(x, y)$ correction reconciles level–set geometry with wealth masses, a formal mapping to a fully discrete (parcel–based) equilibrium is left open. Second, amenities and public goods are omitted; incorporating them would sharpen welfare conclusions. Third, the existence of equilibria need to be proved, and a dynamic framework needs to be employed that allows stability analysis to be done on equilibria that are proved to exist and that allows for programmable simulations to be created so that 2-dimensional visualizations of steady-state spatial equilbria can be produced for different sets of initial conditions and parameter values.